



NAME:
HABIB ID:
SECTION:
INSTRUCTOR:

LINEAR ALGEBRA

SPRING 2026

QUIZ 13 (23rd April 2026)

Max Marks: 10

Time: 10 minutes

Let $\mathbb{R}^2$ have the standard Euclidean inner product.

Use the Gram–Schmidt process to transform the basis

$$u_1 = (1, 1), \quad u_2 = (1, 0)$$

into an **orthonormal basis** of $\mathbb{R}^2$. Sketch both the original basis vectors and the orthonormal basis vectors in the xy -plane.



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Let $\mathbb{R}^2$ have the standard Euclidean inner product. Use the Gram–Schmidt process to transform the basis

$$u_1 = (1, 0), \quad u_2 = (1, 1)$$

into an **orthonormal basis** of $\mathbb{R}^2$. Sketch both the original basis vectors and the orthonormal basis vectors in the xy -plane.



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Solution to Version A (mutiple sol possible) **Step 1:**

$$v_1 = u_1 = (1, 1)$$

Step 2: Projection of u_2 onto v_1 :

$$\text{proj}_{v_1}(u_2) = \frac{u_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \frac{1}{2}(1, 1) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Step 3:

$$v_2 = u_2 - \text{proj}_{v_1}(u_2) = (1, 0) - \left(\frac{1}{2}, \frac{1}{2}\right) = \left(\frac{1}{2}, -\frac{1}{2}\right)$$

Step 4: Normalize

$$\|v_1\| = \sqrt{2}, \quad e_1 = \frac{1}{\sqrt{2}}(1, 1)$$

$$\|v_2\| = \frac{1}{\sqrt{2}}, \quad e_2 = \frac{1}{\sqrt{2}}(1, -1)$$

Final Orthonormal Basis:

$$\left\{ \frac{1}{\sqrt{2}}(1, 1), \frac{1}{\sqrt{2}}(1, -1) \right\}$$

Solution to Version B (mutiple sol possible) **Step 1:**

$$v_1 = u_1 = (1, 0)$$

Step 2: Projection

$$\text{proj}_{v_1}(u_2) = \frac{1}{1}(1, 0)$$

Step 3:

$$v_2 = (1, 1) - (1, 0) = (0, 1)$$

Step 4: Normalize

$$e_1 = (1, 0), \quad e_2 = (0, 1)$$

Final Orthonormal Basis:

$$\{(1, 0), (0, 1)\}$$